

Guideline for the pharmacological treatment of hypertension in adults

Geneva: World Health Organization; 2021

ISBN 978-92-4-003398-6 (electronic version) | ISBN 978-92-4-003397-9 (print version)

(c) World Health Organization 2021

Licence: CC BY-NC-SA 3.0 IGO

This document reproduces the full text of the official WHO guideline for use as a training dataset in the AI Clinical Decision Support Lite Hackathon. Content is reproduced under the CC BY-NC-SA 3.0 IGO licence. Original publication: World Health Organization, Geneva, 2021. Source: <https://www.ncbi.nlm.nih.gov/books/NBK573631/>

Suggested citation

Guideline for the pharmacological treatment of hypertension in adults. Geneva: World Health Organization; 2021. Licence: CC BY-NC-SA 3.0 IGO.

Licence notice

This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; <https://creativecommons.org/licenses/by-nc-sa/3.0/igo>). Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted.

Acronyms and abbreviations

ACEi angiotensin-converting enzyme inhibitor. ARB angiotensin-II-receptor blocker. BB beta-blocker. BP blood pressure. CAD coronary artery disease. CCB calcium channel blocker. CKD chronic kidney disease. CV cardiovascular. CVD cardiovascular disease. DBP diastolic blood pressure. DM diabetes mellitus. ECG electrocardiogram. EML Essential Medicines List. ESRD end-stage renal disease. GDG Guideline Development Group. GRADE Grading of Recommendations Assessment, Development and Evaluation. HCW health care worker (nonphysician). HIC high-income country. HTN hypertension. LIC low-income country. LMIC low- and middle-income country. LVH left ventricular hypertrophy. MACE major adverse cardiovascular event. MI myocardial infarction. MIC middle-income country. NCD noncommunicable disease. PICO population intervention comparator outcome. QALY quality-adjusted life year. RAAS renin-angiotensin-aldosterone system. SBP systolic blood pressure.

Executive summary

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries. Hypertension - or elevated blood pressure - is a serious medical condition that significantly increases the risk of heart, brain, kidney and other diseases. Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control. However, cost-effective treatment options do exist.

In this guideline, the World Health Organization (WHO) provides the most current and relevant evidence-based global public health guidance on the initiation of treatment with pharmacological agents for hypertension in adults. The recommendations target adult, non-pregnant patients who were appropriately diagnosed with hypertension and counselled about life-style modifications.

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, as well as recommendations on intervals for follow up, target blood

pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. The guideline provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy or single-pill combinations, as well as guidance for countries selecting medicines and algorithms for hypertension control for their national guidelines for hypertension management.

The guideline was developed in accordance with the WHO Handbook for Guideline Development. An overview of systematic reviews of the evidence was used to build summary of findings tables according to the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) approach. The Guideline Development Group developed recommendations, considering the certainty of the evidence; the balance between desirable and undesirable effects; resource requirements and cost-effectiveness; health equity; acceptability, patient values and preferences, and feasibility.

Recommendation 1: Blood pressure threshold for initiation of pharmacological treatment

WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥ 140 mmHg or diastolic blood pressure of ≥ 90 mmHg. Strong recommendation, moderate- to high-certainty evidence.

WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130-139 mmHg. Strong recommendation, moderate- to high-certainty evidence.

WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130-139 mmHg. Conditional recommendation, moderate- to high-certainty evidence.

Recommendation 2: Laboratory testing

When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment. Conditional recommendation, low-certainty evidence.

Recommendation 3: Cardiovascular disease risk assessment

WHO suggests cardiovascular disease risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment. Conditional recommendation, low-certainty evidence.

Recommendation 4: Drug classes to be used as first-line agents

For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment: (1) thiazide and thiazide-like agents; (2) angiotensin-converting enzyme inhibitors

(ACEis)/angiotensin-receptor blockers (ARBs); (3) long-acting dihydropyridine calcium channel blockers (CCBs). Strong recommendation, high-certainty evidence.

Recommendation 5: Combination therapy

For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), ACEis/ARBs, and long-acting dihydropyridine CCBs. Conditional recommendation, moderate-certainty evidence.

Recommendation 6: Target blood pressure

WHO recommends a target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities. Strong recommendation, moderate-certainty evidence.

WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD). Strong recommendation, moderate-certainty evidence.

WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease). Conditional recommendation, moderate-certainty evidence.

Recommendation 7: Frequency of re-assessment

WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target. Conditional recommendation, low-certainty evidence.

WHO suggests a follow up every 3-6 months for patients whose blood pressure is under control. Conditional recommendation, low-certainty evidence.

Recommendation 8: Treatment by nonphysician professionals

WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight. Conditional recommendation, low-certainty evidence.

1 Introduction

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries. Blood pressure is the force exerted by circulating blood against the walls of the body's arteries, the

major blood vessels in the body. Blood pressure is written as two numbers. The first number (systolic) represents the pressure in blood vessels when the heart contracts or beats. The second number (diastolic) represents the pressure in the vessels when the heart rests between beats. Hypertension - or elevated blood pressure - is a serious medical condition that significantly increases the risk of diseases of the heart, brain, kidneys and other organs. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control. However, cost-effective treatment options do exist.

A World Health Organization (WHO) guideline dealing specifically with raised blood pressure was last published more than 20 years ago - in 1999 - and is now outdated. In 2007, a comprehensive guideline on cardiovascular risk included some recommendations on hypertension but this also needs revision and updating in the light of new evidence and practice. Guidance is particularly needed on some controversial issues, such as when to start treatment and whether laboratory testing and cardiovascular risk assessment are needed prior to starting treatment.

The WHO Essential Medicines List (EML) identifies all classes of antihypertensive drugs - angiotensin-converting enzyme inhibitors (ACEi), calcium channel blockers (CCB), angiotensin receptor blockers (ARB) and thiazide diuretics - as essential. In June 2019, the EML included single-pill combination medications for hypertension.

1.1 Scope and objectives of the hypertension guideline

The 2021 WHO hypertension guideline aims to provide the most current and relevant evidence-based global public health guidance on the initiation of treatment (with pharmacological agents) for hypertension in adults. The recommendations target the general adult, non-pregnant, hypertensive population.

The objectives of the hypertension guideline are to: provide a blood pressure threshold for the initiation of treatment for hypertension; determine if laboratory tests or cardiovascular risk assessment are required prior to initiation of treatment for hypertension; determine the pharmacological agents with which to initiate treatment; determine the need to initiate treatment with monotherapy, dual therapy, or single-pill combinations; provide targets for blood pressure control in hypertension; provide intervals for follow up for patients with hypertension; and determine how nonphysician health care workers can participate in the management of hypertension.

Nonpharmacological approaches to treatment or prevention of hypertension include: reducing salt intake (to less than 5g daily); eating more fruit and vegetables; being physically active on a regular basis; avoiding use of tobacco; reducing alcohol consumption; limiting the intake of foods high in saturated fats; eliminating/reducing trans fats in diet.

1.2 Target audience

Primary audience: Clinicians and health care providers at all levels of health care. Secondary audience: National NCD/CVD programme managers, health care academics, policy-makers setting practice recommendations, students, and hypertension medicine manufacturers.

2 Method for developing the guideline

2.1 Guideline contributors

In order to develop the hypertension guideline, WHO established three groups: (1) an internal WHO Steering Group to coordinate the guideline development process; (2) a Guideline Development Group (GDG), composed of hypertension physicians, nephrologists, cardiologists, pharmacists, nurses, researchers, academics, policy-makers and representatives of patient groups, to review the evidence and develop recommendations; (3) an External Review Group (ERG), composed of technical experts, representatives of HTN patient groups and ministries of health from low-resource countries, to provide peer review of the guideline.

2.2 Analytical framework and PICOs

An initial GDG meeting was held in Geneva in July 2019 to determine the scope and PICO (population, intervention, comparison, outcome) questions of the guideline. The GDG first developed an analytical framework that demonstrates the impact of interventions on intermediate and final outcomes and displays the order of the key questions to better visualize and place them along the patient-flow pathway.

2.3 Outcome importance rating

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with HTN. The GDG then rated each outcome on a scale from 1 to 9 and indicated whether it considered each outcome critical (rated 7-9), important (rated 4-6) or not important (rated 1-3) for decision-making.

2.4 Reviews of evidence

The WHO Steering Group, with the participation of the GDG, determined the scope of the guideline and identified eleven questions in the format of population, intervention, comparison, and outcomes (PICO) to guide the search for systematic reviews. A systematic search was carried out in PubMed, Embase, The Cochrane Library, and Epistemonikos to identify existing systematic reviews that answered the PICO questions published in 2015 or after. A total of 159 systematic reviews and 17 additional primary studies were included.

2.5 Certainty of evidence and strength of recommendations

The GDG rated the certainty of evidence and developed the recommendations using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach. According to the GRADE approach, the certainty of the evidence can be high, moderate, low, or very low. A strong recommendation is one for which the GDG was confident that the desirable effects of adhering to the recommendation outweigh the undesirable effects. A weak or conditional recommendation is one for

which the GDG concluded that the desirable effects probably outweigh the undesirable effects, but was not confident about these trade-offs.

2.6 Deciding upon recommendations

The GDG met virtually for four sessions in February 2021. Systematic reviews and GRADE tables were presented at the meeting. Formulation of recommendations and their strength were facilitated by the chair and supported by the methodologist, with the aim of achieving unanimous consensus. Full consensus was reached on all recommendations.

2.7 Funding

The development of this guideline was financially supported by the US Centers for Disease Control and Prevention and the World Health Organization.

3 Recommendations

3.1 Blood pressure threshold for initiation of pharmacological treatment

WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥ 140 mmHg or diastolic blood pressure of ≥ 90 mmHg. Strong recommendation, moderate- to high-certainty evidence.

Implementation remarks: Initiation of pharmacological hypertension (HTN) treatment should start no later than four weeks following diagnosis of HTN. If blood pressure level is high (e.g. systolic ≥ 160 mmHg or diastolic ≥ 100 mmHg) or there is accompanying evidence of end organ damage, initiation of treatment should be started without delay.

Evidence and rationale: The GDG reviewed evidence from 14 relevant systematic reviews that summarized data from a large number of randomized trials involving over 120,000 adult participants. On average, treatment was associated with a reduction in deaths and cardiovascular events that ranged from 5 to 10 per 1000 and harms that ranged from 20 to 30 per 1000. The anticipated benefits were large and clearly outweighed the harms. The overall certainty varied from moderate to high, depending on the BP level and agent used.

3.2 Laboratory testing before and during pharmacological treatment

When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment. Conditional recommendation, low-certainty evidence.

Implementation remarks: Suggested tests include serum electrolytes and creatinine, lipid panel, HbA1C or fasting glucose, urine dipstick, and electrocardiogram (ECG). In low-resourced areas or non-clinical settings, where testing may not be possible because of additional costs, and lack of access to laboratories and ECG, treatment should not be delayed, and testing can be done subsequently.

Evidence and rationale: In terms of secondary HTN, various studies suggest a prevalence of 5-10% among patients with a diagnosis of HTN and a higher prevalence of 10-30% among patients with particularly high BP or BP that is resistant to treatment. Overall, the desirable effects of testing were judged to be at least moderate, while the undesirable effects (delay of treatment, cost, incidental findings) were judged to have smaller magnitude.

3.3 Cardiovascular disease risk assessment as guide to initiation of antihypertensive medications

WHO suggests cardiovascular risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment. Conditional recommendation, low-certainty evidence.

Implementation remarks: Most patients with SBP ≥ 140 or DBP ≥ 90 mmHg are high risk and indicated for pharmacological treatment; they do not require cardiovascular (CVD) risk assessment prior to initiating treatment. CVD risk assessment is most important for guiding decisions about initiating pharmacological treatment for hypertension in those with lower SBP (130-139 mmHg).

3.4 Drug classes to be used as first-line agents

For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment: (1) thiazide and thiazide-like agents; (2) angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs); (3) long-acting dihydropyridine calcium channel blockers (CCBs). Strong recommendation, high-certainty evidence.

Implementation remarks: Long-acting antihypertensives are preferred. Examples of indications to consider specific agents include diuretics or CCBs in patients over 65 years or those of African descent, beta-blockers in ischaemic heart disease, ACEis/ARBs in patients with severe proteinuria, diabetes mellitus, heart failure or kidney disease.

Evidence and rationale: Data from 32 systematic reviews were used to derive evidence on benefits and harms of various medication classes. Mortality and major adverse cardiac events (MACE) reduction per 1000 treated people for the various classes were 3 and 14 (low-dose thiazide), 12 and 39 (high-dose thiazide), 23 and 48 (ACEi), 8 and 23 (CCB), 2 and 8 (beta-blockers) and 14 and no data for MACE (ARBs) respectively. The anticipated benefits clearly outweighed the potential harms for thiazide and thiazide-like agents, ACEis/ARBs, and long-acting dihydropyridine CCBs.

3.5 Combination therapy

For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs). Conditional recommendation, moderate-certainty evidence.

Implementation remarks: Combination medication therapy may be especially valuable when the baseline BP is $\geq 20/10$ mmHg higher than the target blood pressure. Single-pill combination therapy improves medication-taking adherence and persistence and BP control.

Evidence and rationale: A large nonrandomized study from Italy (125,635 patients, age 40-85 years) evaluated those who started antihypertensive treatment with one drug vs a two-drug single-pill or free combination. Propensity score adjusted analysis suggests that an initial two-drug single-pill or free combination was associated with significant reductions in the risk of death (20%, 11-28%) and hospitalization for cardiovascular events (16%, 10-21%) compared with initial monotherapy. Adherence, measured as the mean difference in medication possession ratio, was 8-14% higher with a single-pill combination.

3.6 Target blood pressure

WHO recommends a target blood pressure treatment goal of $<140/90$ mmHg in all patients with hypertension without comorbidities. Strong recommendation, moderate-certainty evidence.

WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD). Strong recommendation, moderate-certainty evidence.

WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease). Conditional recommendation, moderate-certainty evidence.

Evidence and rationale: The evidence base consisted of five systematic reviews as well as a review of the SPRINT trial. The desirable effects of lower target BP (per 1000 treated patients) were: a reduction in mortality of 27 (for SBP <120 vs 130-139) and of 7 (for SBP 140/90 vs 150-160); a reduction in cardiovascular mortality of 40 and 6 respectively; and a reduction in stroke of 17 (for SBP <130 vs <140).

3.7 Frequency of re-assessment

WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target. Conditional recommendation, low-certainty evidence.

WHO suggests a follow up every 3-6 months for patients whose blood pressure is under control. Conditional recommendation, low-certainty evidence.

Evidence and rationale: A retrospective population-based cohort study of family-practice clinics in the UK studied 88,756 adults with HTN (1986-2010). Patients with >1.4 months prior to initiation of treatment had a hazard ratio of 1.12 (1.05-1.20) for a major adverse cardiovascular event (MACE) compared to those who started treatment at <1.4 months.

3.8 Administration of treatment by nonphysician professionals

WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight. Conditional recommendation, low-certainty evidence.

Implementation remarks: Community health care workers (HCWs) may assist in tasks such as education, delivery of medications, blood pressure measurement and monitoring through an established collaborative care model. Telemonitoring and community- or home-based self-care are encouraged to enhance the control of BP as a part of an integrated management system.

Evidence and rationale: The evidence base for this question consisted of 11 systematic reviews. Pharmacist-managed care led to better BP control (relative risk 1.44, or 170 more controlled per 1000). Task sharing with pharmacists led to reductions of 8 mmHg SBP and 3.74 mmHg DBP in LICs and MICs, with similar results for nurses (5.34 mmHg lower), dietitians (4.67 mmHg lower), and community HCWs (3.67 mmHg lower).

4 Special settings

4.1 Hypertension in disaster, humanitarian and emergency settings

Hypertension (HTN) is seen in a range of humanitarian crises and disaster settings. There are currently no data regarding target BP or the best antihypertensive agent to treat disaster-related HTN.

Opinion-based recommendation is that the target BP control level should be less than 140 mmHg for SBP and less than 90 mmHg for DBP. Long-acting CCBs are preferred because they are metabolically neutral, and best at reducing BP variability.

4.2 COVID-19 and hypertension

Almost all available evidence suggests that hypertension increases the risk of severe COVID-19. After adjustment for confounders, history of ACEi or ARB use was not found to be associated with increased severity of COVID-19 illness. Most of the world's professional societies either recommend or strongly encourage continuing ACEis/ARBs in COVID-19-infected patients.

4.3 Pregnancy and hypertension

Hypertension, including chronic HTN, gestational HTN, pre-eclampsia, and eclampsia, is a very common medical condition in pregnancy. Up to 10% of pregnancy-related deaths are attributed to HTN. For the pharmacological treatment of HTN in pregnancy, preferred medications include methyldopa, beta-blockers (particularly labetalol), CCBs (particularly nifedipine), and direct-acting vasodilators (particularly hydralazine). There are clear contraindications to the use of ACEis, ARBs, and spironolactone during pregnancy due to direct adverse fetal effects.

5 Publication, implementation, evaluation and research gaps

5.1 Publication

This guideline is available to download from the WHO website. Given that an overview of published systematic reviews was used for the development of the guideline, all reviews are already published and available online.

5.2 Implementation and dissemination

WHO regional and country offices, through their contacts with ministries of health, will encourage implementation at country level. The HEARTS technical package, currently being implemented in 18 countries, will be the platform used to implement and disseminate this guideline.

5.3 Evaluation

WHO will monitor uptake and implementation of the guideline in national policies and programmes by reviewing the number of countries that have adapted or endorsed the guideline nationally.

5.4 Future updating of the guideline

The guideline is expected to be valid for a period of five years. If the evidence base or user needs change before the five-year mark, consideration will be given to producing updates sooner.

5.5 Research gaps

Several research gaps were identified by the GDG according to the theme of the PICOs, including: more evidence on treatment of those in the SBP 130-139 range with diabetes, chronic kidney disease, heart failure, or age 65+; better outcomes data on heart failure and cognitive impairment; more evidence in LICs, MICs and non-North American/European countries; and research on the feasibility and effectiveness of team-based and remote-monitoring care models.

6 Implementation tools

6.1 Guideline recommendations - single-pill combination pathway

Treat adults with BP $\geq 140/90$ mmHg (SBP ≥ 130 mmHg for those with CVD, DM, CKD). Start two-drug combination therapy, preferably in a single-pill combination (ACE/ARB, dihydropyridine CCB, thiazide-like agents). Treatment targets: $<140/90$ mmHg (SBP <130 mmHg for high-risk patients with CVD, DM, CKD). Follow up monthly after initiation or a change in antihypertensive medications until patient reaches BP target. Follow up every 3-6 months for patients with BP under control.

6.1b Guideline recommendations - monotherapy or free-combination pathway

Treat adults with BP $\geq 140/90$ mmHg (SBP ≥ 130 mmHg for those with CVD, DM, CKD). Start with medications from any of the following three classes as an initial treatment: (1) thiazide and thiazide-like agents, (2) ACEi/ARB, and (3) long-acting dihydropyridine CCB. Treatment targets: $<140/90$ mmHg (SBP <130 mmHg for high-risk patients with CVD, DM, CKD). Follow up monthly after initiation or a change in antihypertensive medications until patient reaches target. Follow up every 3-6 months for patients with BP under control.

6.2 Drug- and dose-specific protocols

Algorithm 1 (single-pill combination): Start ARB-CCB at half-maximal doses (example: Telmisartan 40 - Amlodipine 5 mg once a day). Recheck BP in 4-6 weeks. If BP is not at goal, increase ARB-CCB to double dose. If still not at goal, add a thiazide/thiazide-like diuretic at half-maximal dose. If still not at goal, increase the diuretic to double dose. If still not at goal, refer to a specialist. This protocol is contraindicated for women who are or could become pregnant.

Algorithm 2 (monotherapy or free combination): Start a CCB at half-maximal dose (e.g. Amlodipine 5 mg once a day). Recheck BP in 4-6 weeks. If BP is not at goal, increase the CCB to double dose. If still not at goal, add an ARB at half-maximal dose. If still not at goal, increase the ARB to double dose. If still not at goal, add a thiazide/thiazide-like diuretic, then increase it, then refer to a specialist if still not controlled. This protocol is contraindicated for women who are or could become pregnant; neither an ACEi nor ARB should be given to pregnant women.

References

Guideline for the pharmacological treatment of hypertension in adults. Geneva: World Health Organization; 2021. Full reference list, including all 102 cited works spanning systematic reviews, randomized controlled trials (ALLHAT, VALUE, CAMELOT, SPRINT, ACCORD), and health-economic analyses, is available in the original publication at <https://www.ncbi.nlm.nih.gov/books/NBK573631/> and <https://iris.who.int/handle/10665/344424>.

Annex 1: List of contributors

WHO Steering Group members: Bernadette Cappello, Neerja Chowdhury, Gampo Dorji, Jill Farrington, Taskeen Khan, Pedro Ordunez, Steven Shongwe, Slim Slama, Cherian Varghese, Marco Vitoria, Temo Waqanivalu. Guideline Development Group members: Shrish Acharya, Akram Al-Makki, Hind Mamoun Beheiry, Beatriz Champagne, Ugyen Choden, Kenneth Connell, Marie Therese Cooney, Donald DiPette, Nnenna Ezeigwe, Tom Gaziano, Agaba Gidio, Vilma Irazola, Patricio Lopez Jaramillo, Unab Khan, Vindya Kumarapeli, Andrew Moran, Margaret Mswema Silwimba, Brian Rayner, K. Srinath Reddy, Nizal Sarrafzadegan, Apichard Sukonthasan, Paul Whelton, Jing Yu. Methodologist: M Hassan Murad (Mayo Clinic).

Annex 3: Treatment outcomes relevant to hypertension

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with HTN and rated each outcome for its importance to decision-making, ranging from critical (7-9) to important (4-6) to not important (1-3).

Annex 4: PICO questions

Eleven PICO (population, intervention, comparison, outcome) questions guided the search for systematic reviews, covering: BP threshold for treatment initiation; need for laboratory testing; need for CVD risk assessment; monotherapy vs no therapy; monotherapy vs other monotherapy; monotherapy vs combination therapy; choice of combination therapy; single-pill vs free combination; post-treatment target BP; timing of BP re-assessment; and management by physician vs nonphysician provider.